

Lightweight Machine-Learning-Based Diagnosis for Power Electronic Systems Subject to Imbalanced Data

Qingli Deng¹, Student Member, IEEE, Xiaoyun Feng, Shuai Zhao², Member, IEEE, Bin Gou³, Member, IEEE, Huai Wang⁴, Senior Member, IEEE, and Xinglai Ge⁵, Senior Member, IEEE

Abstract—Machine-learning-based fault diagnosis methods have gained extensive applications in power electronic systems. One of the key challenges for field implementation is the rarity of fault history data in practical systems. It leads to an imbalanced ratio between normal and fault state data, causing high accuracy for the majority but severely reducing that of the minority class. To address this issue, a method integrating of oversampling and ensemble learning is proposed in this article. First, the feature extraction and hybrid selection schemes are designed to obtain the most critical features, thereby enhancing the performance of the sampling process. The safe-level synthetic minority oversampling technique is then applied to sample the minority classes to balance the historic database. An ensemble random vector functional link network learning model, which is computationally light, is developed as the diagnostic model to further improve the accuracy. The method is demonstrated by an experimental study on the sensor and open-circuit faults of single-phase rectifiers.

Index Terms—Ensemble learning, fault diagnosis, hybrid feature selection scheme, imbalanced data, safe-level synthetic minority oversampling technique (safe-level SMOTE).

I. INTRODUCTION

A. Background and Motivation

POWER electronic converters play an imperative role in the energy conversion system and are extensively used in many applications, e.g., the grid-interfaced fields of photovoltaic power generation, railway electrical traction transportation, battery charging, and aerospace systems, with high power factor and efficiency [1]. Faults that occur in power devices and sensors may cause immediate system crashes or trigger secondary failures, thereby posing significant safety risks [2]. Therefore,

accurate and real-time fault diagnosis of them is a desirable function for system safety, performance, and maintenance.

In the literature, numerous data-driven fault diagnosis methods have been proposed to deal with the technical challenges posed by multiple fault types and inaccurate mathematical models of the power electronic converters [3]. Fault diagnosis of power electronic converters is commonly regarded as a fault classification problem. Mostly, the diagnostic models are trained by an elaborately designed dataset, which means not only sufficient samples but also a balanced sample distribution for different classes. Unfortunately, the historic dataset is always imbalanced due to the predominant operation of the power electronic system in a normal state for the majority of its operational duration.

As reported in [4], the minority fault samples are prone to be ignored by machine-learning methods because they pay equal attention to each sample. Hence, addressing the imbalanced data problem is important to ensure accurate and reliable fault classification performance, particularly for detecting rare faults that may have critical implications for system operation. Although the challenges of imbalanced data issues have been studied in the theoretical machine learning and some engineering applications for fault diagnosis, the applications in power electronics fault diagnosis are rarely investigated, to our best knowledge.

B. Literature Review

The efforts toward fault diagnosis in power electronic systems, as discussed in [5], can be categorized into: model-based methods (e.g. state-space observer [6], reduced-order observer [7], and mixed logical dynamic model [8]); signal-based methods (e.g. current and voltage signal, [9], [10]); and data-driven methods [11]. Among them, the data-driven fault diagnosis method has gained significant attention in academia due to its independence from system models, load conditions, and signal patterns [12]. It characterizes the nonlinear relationship between fault modes and fault features through a large amount of historical data using machine-learning algorithms. Numerous machine-learning tools in the existing research, including support vector machine (SVM) [13], Bayesian network [14], neural network [15], and random forest (RF) algorithm [16], are utilized to build the classification model.

In power electronic converters, accurate fault diagnosis is of paramount importance, and it relies on effectively capturing

Manuscript received 31 July 2023; revised 2 December 2023; accepted 17 January 2024. Date of publication 25 January 2024; date of current version 18 April 2024. This work was supported in part by the High-Speed Railway Joint Funds of the National Natural Science Foundation of China under Grant U1934204 and in part by the State Scholarship Fund of China Scholarship Council (CSC). (Corresponding author: Xiaoyun Feng.)

Qingli Deng, Xiaoyun Feng, Bin Gou, and Xinglai Ge are with the Ministry of Education Key Laboratory of Magnetic Suspension Technology and Maglev Vehicle, Southwest Jiaotong University, Chengdu 610031, China (e-mail: qinglideng@my.swjtu.edu.cn; fengxy@home.swjtu.edu.cn; bingou@swjtu.edu.cn; xlge@swjtu.edu.cn).

Shuai Zhao and Huai Wang are with the AAU Energy, Aalborg University, 9220 Aalborg, Denmark (e-mail: szh@energy.aau.dk; hwa@energy.aau.dk).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/JESTIE.2024.3358729>.

Digital Object Identifier 10.1109/JESTIE.2024.3358729

and representing distinctive fault features related to changes in voltage or current, which directly reflect the system's physical properties. To achieve this, feature extraction and selection algorithms play a key role in the initial step. Notable works by the authors in [14] and [15] have utilized fast Fourier transformation (FFT) to extract frequency-domain features from fault signals originally in the time domain, followed by critical feature selection using principal component analysis (PCA) and ReliefF algorithms. In addition, enhancing the classification capability is crucial for accurate fault identification. For instance, the ensemble learning technique presented in [17] is designed to augment diversity within the learning algorithms, thereby significantly improving the accuracy of identifying various types of sensor faults. These efforts collectively contribute to the pursuit of high accuracy in the fault diagnosis model for power electronic converters.

Besides, fault diagnosis in power electronic systems typically necessitates real-time implementation on embedded hardware platforms, which highlights the requirement for a computationally light diagnostic algorithm. In the case of open-circuit faults in rectifiers, Deng et al. [18] comprehensively consider feature vector design and algorithm parameter adjustment to tackle the issue of online lightweight computation. Nowadays, deep-learning-based diagnosis methods like sparse autoencoder [19], convolutional neural network (CNN) [20], and deep sparse filtering network [21], have been also employed in the insulated-gate bipolar transistor (IGBT) open-circuit or sensor faults in a power electronic system, their verification platforms mainly rely on computers. However, the real-time performance and feasibility on embedded hardware have been rarely explored. To sum up, fault diagnosis in the field of power electronics is expected to be implemented for real-time detection and achieve high accuracy.

Regarding the issue of fault diagnosis with imbalanced data, it has received significant attention and research in various domains, such as disease diagnosis [22], high-voltage circuit breakers diagnosis [23], machine diagnosis [4], bearing fault diagnosis [24], and side-channel attacks detection [25]. Among them, the efforts to improve the diagnostic accuracy of the minority class are made from the perspective of data and algorithm [26]. Generally, data-level methods include various resampling techniques, such as oversampling [22], [23] and undersampling methods [4], [26]. At the algorithm level, representation learning, such as autoencoder [24], transfer learning [24], deep learning [25], and ensemble learning [27] are applied to explore the details of the datasets. Nevertheless, no study is found in fault diagnosis for power electronics applications specifically considering imbalanced data. While effective solutions, such as data resampling and ensemble learning techniques, exist for imbalanced data, their effectiveness may vary due to different data distributions, domain-specific prior knowledge, constraints, and system requirements.

For instance, there are differences in the application of data resampling techniques for different data types and distributions. Generally, the oversampling method, which augments data by generating minority samples, is preferred over the undersampling method since it balances data by discarding majority samples. The synthetic minority over-sampling technique (SMOTE),

as one of the most widely applied methods, generates new synthetic minority samples based on the K-nearest neighbor (KNN) algorithm [28]. Since the original data distribution is a critical impact factor in deciding the newly generated sample quality, the improved variants of borderline-SMOTE [29], safe-level SMOTE [30], and adaptive synthetic sampling approach [31] are introduced. Although the new samples contribute little to determining the decision boundary and may lead to overlap problems, the advantages of the aforementioned sampling techniques are simplified to generate new samples around the minority samples, further balancing the dataset. As such, dealing with data under the specific fault patterns and characteristics of power electronic systems, to obtain high-quality features and effectively sample new data, presents the first challenge.

In addition, when considering data imbalance, the second challenge is to ensure both model accuracy and feasibility for real-time diagnostic requirements while applying ensemble learning models in fault diagnosis for power electronics systems. With its powerful modeling capability, deep learning techniques, such as autoencoder, transfer learning, and ensemble learning, are promising in the field of bearing fault diagnosis and security evaluation with imbalanced data. In [24], an integrated multitasking intelligent algorithm of autoencoder and transfer learning is proposed for unsupervised bearing fault diagnosis under an imbalanced sample. As it can be seen, deep learning is a promising solution in bearing fault diagnosis and it also performs well in the field of security evaluation [25]. However, such a computationally extensive method cannot be implemented on the existing computational units in power electronics systems, which can be used for lightweight applications like control and protection. By contrast, ensemble learning is flexible because the number of the base classifier can be adjusted. Consequently, the ensuing challenge is to reduce the number of ensemble classifiers while maintaining accuracy.

In conclusion, directly implementing existing solutions from other domains to power electronics is not applicable since these unique challenges necessitate tailored solutions that meet the specific demands of fault diagnosis for power electronic converters subject to imbalanced data.

C. New Contributions

To address the imbalanced data challenge in the field of power electronics, this article proposes a method that integrates oversampling and ensemble learning-based fault diagnosis approaches. The proposed implementation pipeline is demonstrated by a case study on a single-phase pulswidth modulation (PWM) rectifier. The main contributions of this article are summarized as follows.

- 1) An integrated approach of oversampling and ensemble learning-based diagnosis method is proposed to address the challenge of imbalanced data.
- 2) The hybrid feature selection is employed to identify the most crucial features, which not only guarantees the quality of the oversampling process but also reduces the feature dimensionality, thereby lowering the complexity of the diagnostic model.
- 3) A fast learning network and its ensemble learning structure with a decision-making strategy are developed to formulate

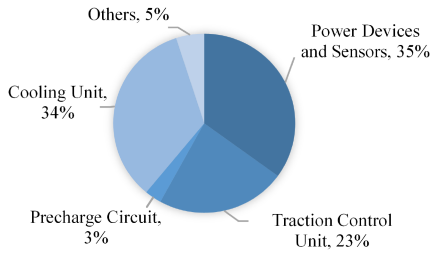


Fig. 1. Industrial statistics of failure rate for traction converter systems [32].

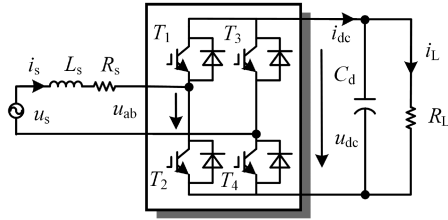


Fig. 2. Structure of the single-phase PWM rectifier.

the diagnostic model, facilitating high accuracy and real-time implementation in power electronic systems.

The remainder of this article is organized as follows. The problem description for the diagnosis of the single-phase PWM rectifier is presented in Section II. Section III provides the details of the proposed methods and implemented pipeline. Then, the proposed scheme is verified by the experimental results in Section IV. Finally, Section V concludes this article.

II. PROBLEM DESCRIPTION

According to an industrial survey of traction converter systems, as shown in Fig. 1, the power devices and sensor failures have the highest proportion [32]. Specifically, 21.5% of these failures are attributed to the single-phase PWM rectifier, while 13.5% are related to the traction inverter. Therefore, the single-phase PWM rectifier is selected as a case study to illustrate the issue of data imbalance and propose tailored solutions for power electronic systems.

The single-phase PWM rectifier is always utilized as a front-end part for the ac–dc–ac traction drive system or a submodule of the cascaded rectifier system, as illustrated in Fig. 2. Here, u_s , i_s , u_{ab} , R_s , and L_s represent the ac-side voltage, current, input voltage of the rectifier, equivalent resistance, and inductance, respectively. T_1 , T_2 , T_3 , and T_4 denote four IGBTs. In addition, u_{dc} , i_{dc} , i_L , C_d , and R_L represent the dc-side voltage, current, load current, capacitor, and equivalent load, respectively. Note that power device open-circuit faults and sensor faults in power electronics are extensively studied. Therefore, normal and four fault types, including T_1 open-circuit fault, T_3 open-circuit fault, current sensor shift, and gain fault are considered as a case study in this work, which are labeled as 0, 1, 2, 3, and 4, respectively.

To illustrate the impact of imbalanced data and emphasize the importance of addressing this issue, a test is conducted by using historical data from the single-phase PWM rectifier. Several classic machine-learning models are trained using data

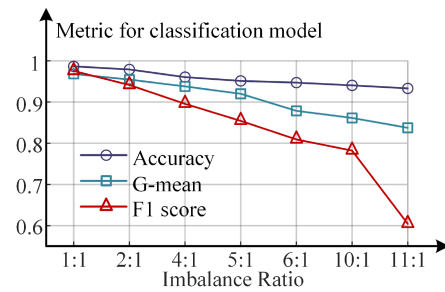


Fig. 3. Accuracy G-mean and F1 score for various imbalanced ratios.

characterized by different imbalance ratios between normal and fault states, and then, evaluated by the same test data. When dealing with imbalanced data, accuracy is not an appropriate metric for evaluating the performance of diagnostic models. This is because accuracy tends to be biased toward the majority class, which is often the normal state. In contrast, G-mean [23] and F1 score [22] are commonly used for imbalanced data as they provide separate measures of classification performance for each class. Hence, the evaluation results in terms of accuracy, G-mean, and F1 score for various imbalance ratios are shown in Fig. 3. It clearly demonstrates that as the ratio between majority and minority fault classes increases, both G-mean and F1 score decrease significantly. This indicates the difficulty in accurately classifying the minority fault instances in the presence of imbalanced data.

Therefore, addressing the imbalanced data issue in fault diagnosis for power electronic converters is of utmost importance. The main objective of this study is to develop an effective solution for improving the performance and robustness of fault diagnostic models in power electronic systems. Specifically, the focus is on improving the offline training accuracy while meeting the real-time requirements for online fault diagnosis.

III. DIAGNOSTIC STRATEGY

A. Proposed Method

As illustrated in Fig. 4, the proposed method for the fault diagnosis with imbalanced data consists of four phases, i.e., feature extraction and selection, imbalanced data resampling, ensemble learning, and real-time fault diagnosis. The implementation pipeline can be divided into two stages: offline training process and real-time application.

In the offline training stage, the aim is to balance the historical dataset by sampling the minority class fault data. To enhance the quality of the newly obtained data, two steps are taken out before executing the sampling algorithm. First, the frequency-domain feature components are extracted serving as the sampled signals, thereby enhancing the fault feature diversity. Then, to prevent the sampling of irrelevant feature data, the most critical features are selected based on their significance and correlation, which could further reduce the dimensionality of feature vectors. Afterward, an oversampling method is devised to generate minority faulty samples randomly. It is worth highlighting that the well-selected fault features are the premise of accurate new data sampling.

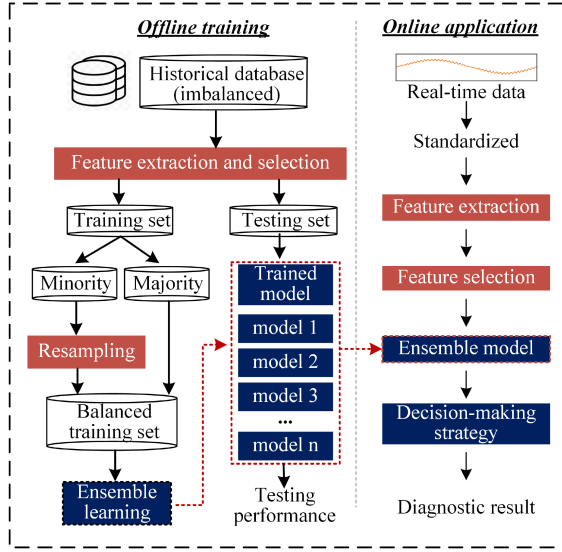


Fig. 4. Offline model training and online diagnostic application process.

Second, based on the balanced training dataset, a random vector functional link network (RVFL) [18] is employed to map the knowledge relationship between the selected features and fault labels. To further improve the diagnostic accuracy, an ensemble learning structure with diversified RVFL models is developed to recognize the fault pattern.

At the online application stage, the frequency features are first extracted from the real-time sampling signals with standardized values. The critical features are selected by following the offline feature selection results. Then, they are inputted into each trained RVFL-based diagnostic model to calculate the fault type, respectively. Besides, to satisfy the needs of diagnostic model online computing, the parameters of RVFL-based ensemble learning models are tuned as simply as possible. Finally, based on the different model outputs, a decision-making strategy scheme inspired by the voting strategy is designed to decide the final fault output. The details are elaborated as follows.

B. Feature Extraction and Selection

In the designed cases, the grid-side current is selected as the diagnostic signal based on the fault characteristic analysis. And it exhibits different characteristics in the time domain and frequency domain. Considering the requirements of extracting the frequency-domain component, the grid-side current in a half cycle of a fundamental period is regarded as the original data. To avoid the dimensional effect, the grid-side current is normalized by dividing the reference current, which is calculated as

$$i_s^* = \frac{i_s}{i_{ref}} = i_s \cdot \frac{u_s R_L}{u_{dc}^2} \quad (1)$$

where the i_s^* is the unitization value, i_s is the measured value, and i_{ref} is reference current, which is computed by the power balancing equation.

The FFT is applied to generate the spectrum features in the frequency domain, increasing the feature diversity. Frequency-domain features can provide detailed information on the

frequency components and energy distribution of the signal, which in turn helps to analyze key features and patterns in the signal. Afterward, the feature vector is created by combining the raw current data and the frequency-domain data, which is expressed as

$$X = (i_1, i_2, \dots, i_n, F_1, F_2, \dots, F_n) \quad (2)$$

where the $i_1, i_2, \dots, i_n$ are the original grid-side current data in a half cycle of a fundamental period, and the $F_1, F_2, \dots, F_n$ denotes the magnitude value of the complex data in the frequency domain.

To avoid the effect of unimportant and irrelevant features on the resampling process, it is important to evaluate the importance and correlation of the features, since the quality of the original data determines that of the new sampled data. As such, the features are selected based on importance, relevance, and redundancy evaluation. A well-selected feature subset not only facilitates the safe-SMOTE sampling procedure since the sampling process requires distance calculations between data points but also lowers computational complexity by reducing the input vector dimensionality for the diagnostic model.

Generally, filter-based and wrapper-based feature selection methods are two typical methods, where filter-based methods evaluate the merits of features based on heuristics, whereas wrapper-based methods select the features based on the performance of a machine-learning algorithm [33]. In this study, the filter-based feature selection methods of ReliefF [17] and minimum redundancy maximum relevance (mRMR) [34] are hybrid utilized for feature ranking. The wrapper-based method based on common classifiers is then trained to select feature subsets.

The ReliefF is a feature-weighting algorithm, which assigns different weights to features based on the correlation between category and feature. Beginning with randomly taking out a sample R , the KNN samples are found from the same sample sets. The KNN samples are discovered from the different sample sets. Then, each feature weight is updated to address the multi-classification problem. The weight of feature A is expressed as

$$W(A) = W(A) - \sum_{j=1}^k \text{diff}(A, R, H_j) / (m \cdot k) + \sum_{C \notin \text{class}(R)} \left[\frac{p(C)}{1 - p(\text{class}(R))} \cdot \sum_{j=1}^k \text{diff}(A, R, M_j(C)) \right] / (m \cdot k) \quad (3)$$

where C is a class label, $p(\cdot)$ is the prior probability of a class, and k is a predefined parameter. The $M_j(C)$ describes the j th nearest neighbor sample in class C , while $\text{diff}(A, R_1, R_2)$ represents the difference between samples R_1 and R_2 , shown as

$$\text{diff}(A, R_1, R_2) = \frac{|R_1[A] - R_2[A]|}{\max(A) - \min(A)}. \quad (4)$$

The key idea of the ReliefF algorithm is to evaluate the importance of features by comparing the differences between samples of different categories and samples of the same category. The algorithm assumes that features with greater variance, are more likely to be important for differentiating the classes of samples. By continuously updating the weights of features, the ReliefF algorithm is able to find the most discriminative subset of features, thereby improving classification performance and reducing feature dimensions. Generally, the greater the feature weight, the stronger the classification ability of the feature. As a result, the feature would be eliminated if its feature weight is smaller than the weight threshold, which is set as zero in this work.

Subsequently, the mRMR is applied to further assess the redundancy and relevance between features and classes. The use of mRMR is to maximize the relevance between features and categorical variables while minimizing the redundancy between features. The metric of mRMR is obtained by mutual information formulated as

$$I(x; y) = \iint p(x, y) \log \frac{p(x, y)}{p(x)p(y)} dx dy \quad (5)$$

where x and y are two random variables, and I represents the mutual information of them. $p(x)$, $p(y)$, and $p(x, y)$ are the probability density functions.

In terms of mutual information, the feature selection process is to find a subset S with m features x_i , which has the biggest dependency on the target class C . This rule is called max-dependency, following the form

$$\max D(S, C), D = I(\{x_i, i = 1, \dots, m\}; C). \quad (6)$$

The max-dependence criterion is hard to implement since the mutual information cannot be estimated accurately when the sample number is large. Thus, max-relevance is introduced to search features by using (7), which approximates in (6) with the mean value of all mutual information values between individual feature x_i and class C as follows:

$$\max D(S, C), D = \frac{1}{|S|} \sum_{x_i \in S} I(x_i; C). \quad (7)$$

Furthermore, to remove the relevance but redundancy feature, the following min-redundancy condition can be added to select mutually exclusive features, which is expressed as

$$\min R(S), R = \frac{1}{|S|^2} \sum_{x_i, x_j \in S} I(x_i, x_j). \quad (8)$$

Combined with the aforementioned two constraints together, the mRMR is achieved by solving the feature set

$$\max \Phi(D, R), \Phi = D - R. \quad (9)$$

As a result, the mRMR algorithm can effectively select features with high correlation with the target variable and low redundancy between each other, thereby reducing the feature dimension. To sum up, the ReliefF evaluates the relationship between features and categories by assigning weights without considering the correlation between features. While the mRMR assesses both the correlation between features and categories

Algorithm 1: Feature Extraction and Hybrid Feature Selection With ReliefF and mRMR.

Input Current signal with labels C of half a fundamental period

Output Subset of features with labels C

1: Normalize current signal $(i_1, i_2, \dots, i_n)$ using (1).

2: Extract frequency-domain features $(F_1, F_2, \dots, F_n)$.

3: Build the original feature sequence as (2).

4: Calculate feature weights using the ReliefF algorithm.

4-1: Initialize feature weights with zeros.

4-2: **For** $i = 1$ to m

a. Select a random instance from the dataset (R_i) .

b. Find k nearest instances with the same class (H_j) .

c. Find k nearest instances with a different class (M_j) .

end

4-3: **For** each feature A

a. Calculate (4) between sample R_1 and R_2 .

b. Update $W(A)$ by (3).

end

4-4: Rank features based on their weights.

5: Remove features with weights below a specified threshold.

6: Rank the retained feature subset using the mRMR.

6-1: Initialize the selected features list as empty.

6-2: **For** each feature not in selected features:

a. Calculate max-relevance by (7) between the feature and label.

b. Calculate min-redundancy by (8) between features.

c. Calculate the mRMR score by (9).

end

6-3: Rank features based on the mRMR score.

7: Train classifiers with different feature subsets.

8: Return the feature subset with the highest train accuracy.

and the redundancy between features, the contribution of each selected feature to the classifier is uniform. Therefore, by combining the ReliefF with the mRMR algorithms, the fault features are ranked. Based on this, several classifiers are trained by incrementally introducing feature subsets, and then, the best feature vector subset is selected with the highest training accuracy.

The pseudocode for the aforementioned feature extraction and hybrid feature selection with ReliefF and mRMR is presented as follows.

C. Oversampling Technique

The resampling technique is generally applied to overcome the challenge of imbalanced data issues including the oversampling or undersampling methods. Commonly, the oversampling methods are more suitable for the power electronic system with a small dataset since the undersampling may lose some important information, which jeopardizes the testing accuracy. The reported oversampling techniques, e.g., synthetic minority oversampling technique (SMOTE) [28], borderline-SMOTE [30], safe-level SMOTE [29], adaptive synthetic sampling approach

Algorithm 2: Safe-Level SMOTE.

Input Minority class data D , sl_p and sl_r , Sampling number l

Output Balanced dataset with synthetic samples

- 1: Calculate the value of α :
 - If** ($sl_r = \infty$ and $sl_p = 0$)
 - a: does not generate positive synthetic instance.
 - For** $i = 1$ to l
 - else if** ($sl_r = \infty$ and $sl_p \neq 0$)
 - b: set the α to 0.
 - else if** ($sl_r = 1$)
 - c: set the α to a random number between 0 and 1.
 - else if** ($sl_r > 1$)
 - d: set the α to a random number between 0 and $1/sl_r$.
 - else if** ($sl_r < 1$)
 - e: set α to a random number between $1 - 1/sl_r$ and 1.
 - end**
- 2: Interpolate the new synthetic sample by (10).
- 3: Combine the D with the synthetic samples.
- 4: Return the balanced dataset.

(ADASYN) [31], generate synthetic samples randomly by selecting the nearest minority neighbors of minority samples according to

$$x_{\text{new}} = x_i + \alpha (x_{zi} - x_i) \quad (10)$$

where the x_i belongs to the minority class samples, x_{zi} is a neighbor of x_i , and α is a random number in the range of $[0, 1]$.

Among them, the safe-level SMOTE is the preferred method due to its capability to assign a safe level to each instance before generating synthetic samples. This approach introduces a constraint, ensuring the quality of the sampled data. Consequently, new synthetic instances are exclusively generated in safe regions, positioned in proximity to the highest safe level. The calculation method for safe level is as follows.

1) Defined that D is a set of all original minority positive instances, computing the KNNs for each sample p in D , and randomly selecting one of them, named as n .

2) sl_p is equal to the instance number of KNNs p in D , while sl_n is equal to the instance number of KNNs n in D .

3) If sl_n is not equal to 0, that is, there are positive instances among the nearest neighbors of n , then the safe level ratio sl_r is equal to the quotient between sl_p and sl_n . Otherwise, if sl_n is equal to 0, set the sl_r to infinity.

Based on the safe-level value, the pseudocode for safe-level SMOTE is shown as follows.

D. Ensemble Learning and Decision-Making Strategy

It is well-known that the fault diagnosis of power electronic systems is time-critical and requires real-time decision making. Therefore, it is crucial to design a computation-light algorithm and framework that not only can be implemented on embedded computation units but also obtain high efficiency and high accuracy. The RVFL [36] was originally proposed for the following advantages.

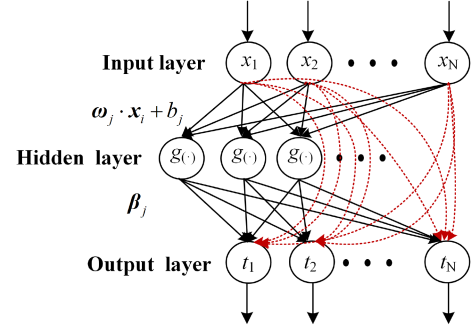


Fig. 5. Structure of the RVFL network.

1) *Fast train speed*: RVFL employs random initialization of all weights and biases between the input layer and hidden layer nodes. These parameters remain fixed and do not require to be tuned during training, resulting in superior training speed.

2) *Flexible regularization*: The direct connection between the input layer and the output layer serves as an efficient and simple regularization technique, preventing RVFL networks from overfitting.

Fig. 5 shows the structure of the RVFL network, a generalized single hidden layer feedforward neural network, including three layers (i.e., the input layer, the hidden layer, the output layer) and the direct link between the input and output layer. For a dataset consisting of $\{(x_i, y_i) \mid x_i \in \mathbb{R}^n, y_i \in \mathbb{R}^m, i = 1, \dots, N\}$, x_i is a feature vector, and y_i is the corresponding output label, the RVFL network can be formulated as

$$f(\mathbf{X}) = \sum_{j=1}^J \beta_j g(\omega_j^T \mathbf{X} + b_j) + \sum_{j=J+1}^{J+N} \beta_j x_j \quad (11)$$

where ω_j and b_j are the weights and biases of the hidden nodes between the input layer and hidden layer, β_j is the output weights, and $g(\cdot)$ is the activation function. Besides, J is the number of hidden nodes, while N is the number of input nodes.

The RVFL network also can be written in a matrix form as

$$\mathbf{H}\beta = \mathbf{Y} \quad (12)$$

where β is an output weight vector, $\mathbf{Y}$ is a label matrix, and matrix $\mathbf{H}$ is equal to $[\mathbf{H}_1 \mathbf{H}_2]$, including the input matrix $\mathbf{H}_1$ and the hidden layer matrix $\mathbf{H}_2$, which are denoted as follows:

$$\mathbf{H}_1 = \begin{bmatrix} x_{11} & \cdots & x_{1N} \\ \vdots & \ddots & \vdots \\ x_{N1} & \cdots & x_{NN} \end{bmatrix}$$

$$\mathbf{H}_2 = \begin{bmatrix} g(\omega_1 \cdot x_1 + b_1) & \cdots & g(\omega_J \cdot x_1 + b_J) \\ \vdots & \ddots & \vdots \\ g(\omega_1 \cdot x_N + b_1) & \cdots & g(\omega_J \cdot x_N + b_J) \end{bmatrix}. \quad (13)$$

For the RVFL, the output weights β can be obtained by solving (12) by using the Moore–Penrose pseudoinverse [35] as follows:

$$\beta = \mathbf{H}^\dagger \mathbf{Y} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{Y}. \quad (14)$$

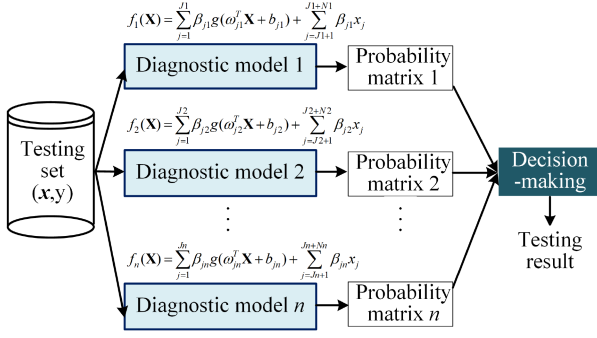


Fig. 6. Testing process with ensemble models.

The rectifier fault diagnosis is a multiclassification task in this study, suggesting that there are multiple output nodes of the RVFL network. Typically, the predicted output corresponds to the node with the highest value among the multiple output nodes, from which the corresponding fault label can be derived. This decision process is given as

$$\text{label}(x) = \arg \max_{i \in \{1, 2, \dots, m\}} f_i(x) \quad (15)$$

where m and $f_i(x)$ are the number of classification labels and the i_{th} output node value, respectively.

Note that the input layer weight is generated randomly, leading to inherent stochasticity in the model. As a result, RVFL often lacks consistency and stability, which can impact its accuracy. To address this limitation and improve accuracy, this article proposes an ensemble learning approach. By combining multiple RVFL models, the ensemble learning process aims to enhance the overall diagnostic performance. Ensemble learning accomplishes learning tasks by building and combining multiple classifiers, which are generated by randomly setting different parameters. Hence, n RVFLs are trained and verified by fivefold hierarchical cross validation.

Generally, the final output of the ensemble classifiers is decided by the supermajority voting method, but it cannot obtain a specific probability of each fault pattern. Thus, a decision-making strategy based on mathematical transformation is formulated. In this case, the softmax function is utilized to convert the classification model output to a probability matrix, ranging from 0 to 1, which is shown as

$$\text{softmax}(t_i) = \frac{\exp(t_i)}{\sum_{j=1}^N \exp(t_j)}. \quad (16)$$

Consequently, n probability matrices for n RVFLs are calculated. The decision-making strategy is to average the probability for each fault pattern and the maximum value is used to decide the final diagnostic output. To sum up, the testing process is implemented based on test datasets, which is illustrated in Fig. 6. And the pseudocode for RVFL-based ensemble learning is presented as follows.

IV. EXPERIMENTAL DEMONSTRATION AND ANALYSIS

The performance of the proposed method is demonstrated by using a single-phase PWM rectifier platform, as shown in Fig. 7.

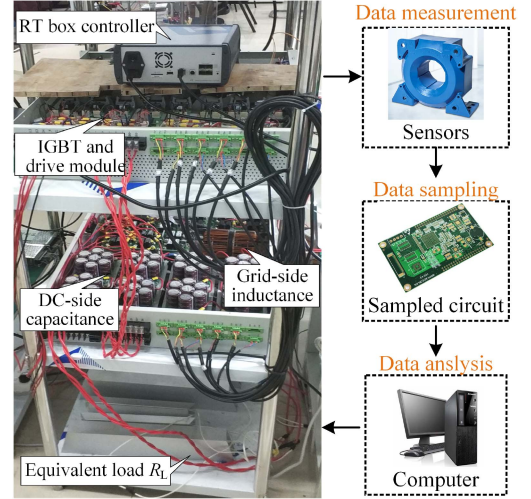


Fig. 7. Experimental data generation and data analysis setup.

TABLE I
PARAMETERS OF SINGLE-PHASE PWM RECTIFIER

Parameter	Value	Parameter	Value
Grid voltage $u_g/\text{V(RMS)}$	50-70	DC-link capacitor C/mF	2.8
Grid frequency f_g/Hz	50	Rated equivalent load R_L/Ω	20/40
Equivalent filter inductor L_s/mH	3	Switching frequency f_i/kHz	1
Parasitic resistor R_s/Ω	0.1	Sampling frequency f_s/kHz	20
Reference DC-link voltage U_{dc}/V	120	Control cycle $T_c/\mu\text{s}$	50

Algorithm 3: RVFL-Based Ensemble Learning.

Input Train dataset, Classifiers number n , Test dataset
Output Ensemble predictions for the test dataset
1: For $i = 1$ to n :
 1-1: Create a RVFL network with initial parameters.
 1-2: Train the RVFL by (11)–(15).
 1-3: Parameter tuning to obtain a high train accuracy.
 end
2: For $i = 1$ to n :
 2-1: Use the trained model to make predictions by (11).
 2-2: Transform to a probability matrix by (16).
 end
3: Average the probability for each fault class.
4: Decision making by majority voting.
5: Return the fault class with the maximum probability value.

A. Data Acquisition

The rectifier system specifications are listed in Table I. To collect data in various operating cases, the grid-side current signal of the normal state, along with four types of electrical faults, i.e., T_1 open-circuit fault, T_3 open-circuit fault, current sensor gain fault, and shift fault, are acquired. The data acquisition is performed separately under the conditions of varying grid-side voltage

and load variations. Considering the rated power limitation, the grid-side voltage amplitude ranges from 50 to 70 V, while the load varies between 20 and 40 Ω . A dataset is constructed with an imbalanced ratio of 4:1:1:1:1, comprising 160 samples acquired in the normal state and 40 samples collected in each fault state. Subsequently, the following tests are conducted using the original imbalanced dataset, consisting of 320 samples.

B. Test Cases Design

To demonstrate the effectiveness of key steps in the proposed method, comprehensive analysis and ablative studies are conducted by the following tests.

1) *Test I*: This test involves the extraction and selection of a low-dimensional and high-accuracy feature subset. It utilizes a hybrid feature selection process to identify the most relevant features from the raw imbalanced data. This test involves feature extraction and obtaining a low-dimensional, high-accuracy feature subset using hybrid feature selection from raw imbalanced data.

2) *Test II*: In this test, various data resampling techniques are utilized to address the class imbalance issue and create a balanced dataset. The performance of these techniques is evaluated and compared by the evaluation metrics including accuracy, F1 score, and G-mean.

3) *Test III*: This test focuses on comparing the classification performance of common learning algorithms with an ensemble RVFL network. The goal is to assess the superiority of the proposed method in terms of classification performance through ensemble learning.

4) *Test IV*: The final test demonstrates the practical applicability of the proposed method in real-world scenarios. It focuses on implementing online diagnosis using real-time computing, with online sampling data.

Together, these four tests showcase the capabilities and advantages of the proposed method across different aspects of the diagnostic process.

C. Test I Feature Subset Selection

As illustrated in Fig. 8, it begins by sampling the current waveform within a half fundamental period. The FFT is then used to generate the frequency components. To determine the importance of each feature, a weight is computed using the ReliefF algorithm. The study initially filters out features with weights smaller than zero, retaining 100 features. Then, the mRMR method is applied to assess the redundancy and correlation among these features, resulting in a reranking and final selection of the 100 features.

To confirm the effectiveness of the feature selection process, four feature vectors, including the raw data, the data ranked by mRMR, ReliefF, and a combination of both, are initially acquired. The feature vectors are then introduced sequentially to train and test classifiers, enabling computation of the test accuracy. In addition, several widely used classification algorithms, including KNN, SVM, RF, decision tree (DT), and RVFL, are employed to mitigate the reliance on a single classifier's

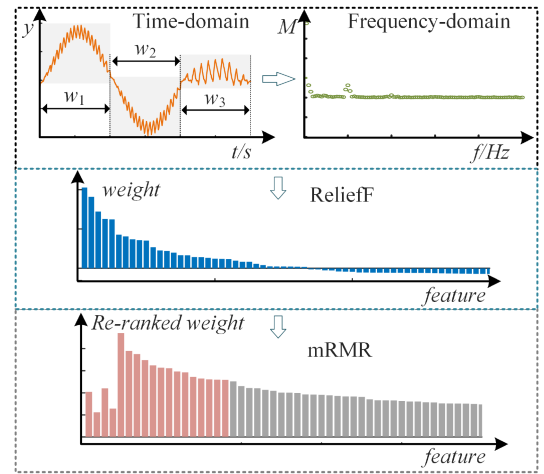


Fig. 8. Generation process of feature subset.

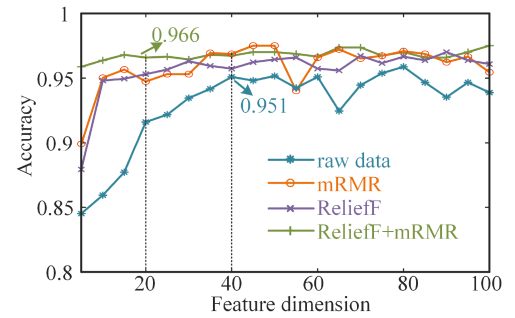


Fig. 9. Test accuracy of classifiers with varying number of features.

characteristics. The average test accuracy for these feature vectors is depicted in Fig. 9.

As observed, all of them achieve an accuracy exceeding 95.1% when the feature dimension exceeds 40. This implies that without any feature selection considerations, high-dimensional features demonstrate acceptable effectiveness. However, when the most crucial features are filtered through ReliefF and mRMR algorithms, the accuracy reaches 96.6% even with a feature dimension of 20. This comparison highlights the advantages and necessities of the ReliefF and mRMR algorithms. Consequently, the feature subset comprises the top 20-D features.

D. Test II Effectiveness of Safe-Level SMOTE

Using the selected feature vector subset from Section IV-C, four overresampling approaches, including SMOTE, borderline SMOTE, safe-level SMOTE, and ADASYN, are used to generate new samples. The effectiveness of the oversampling method is assessed using three imbalanced ratios of 4:1:1:1:1, 8:1:1:1:1, and 12:1:1:1:1, for illustrative purposes. These datasets are created by randomly selecting normal samples and an appropriate number of samples from each type of fault. To evaluate the quality of newly sampled data obtained from different sampling algorithms, various classification algorithms, including KNN, DT, RF, extreme learning machine (ELM) and RVFL, are trained and tested on the same testing dataset. Their performance are

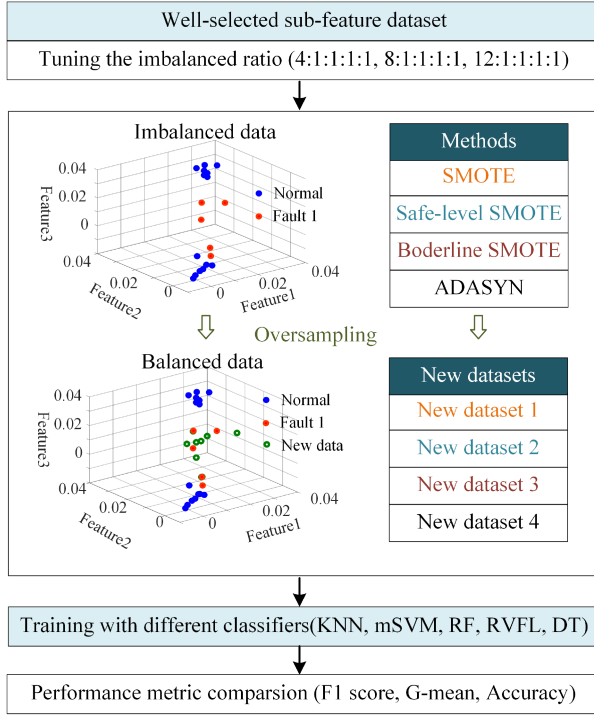


Fig. 10. Verification process of the oversampling method.

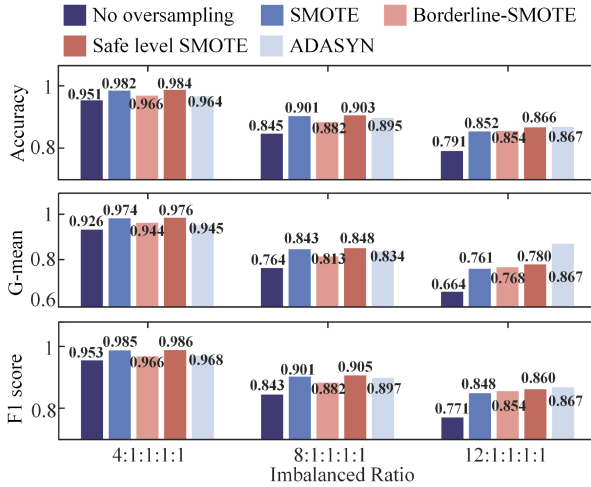


Fig. 11. Metrics for oversampling methods at different imbalanced ratios between normal and four fault types.

assessed by comparing different metrics, including testing accuracy, F1 score, and G-mean. The detailed verification process is depicted in Fig. 10.

To compare the performance of four oversampling algorithms with different imbalanced ratios, the average diagnostic results for five classifiers are calculated based on test accuracy, F1 score, and G-mean. Accordingly, the average metrics value of five diagnostic models are presented in Fig. 11. Comparing these results with no oversampling, the safe-level SMOTE showed the highest improvement in the G-mean metric, with an average increase of 6.1% $((3.2\% + 6.2\% + 8.9\%) / 3 = 6.1\%)$. Similarly, the improvement of SMOTE, Borderline-SMOTE, and ADASYN are 5.5%, 4.4%, and 5.4%, respectively.

TABLE II
COMPARISON OF CLASSIFICATION MODELS UNDER IMBALANCED RATIO OF 4:1:1:1:1

Metric	Train accuracy	Test accuracy	Test time	F1 score	G-mean
KNN	0.9818	0.9762	0.0313	0.9566	0.9753
SVM	0.7030	0.8095	0.1094	0.7040	0.7404
DT	0.9848	0.9333	0.0469	0.9253	0.9556
RF	0.9848	0.9619	0.1719	0.9571	0.9695
ELM	0.9515	0.9286	0.0781	0.9151	0.9497
BPNN	0.9848	0.9571	0.0156	0.9405	0.9721
CNN	0.9909	0.9714	1.2625	0.9667	0.9791
RVFL	0.9758	0.9667	0.0156	0.9619	0.9768
RVFLs	0.9909	0.9810	1.2188	0.9751	0.9838

As for testing accuracy, the safe-level SMOTE algorithm achieved the highest testing accuracy improvement with a 5.5% increase, while others are 4.9%, 3.8%, and 4.6%. Although it is inferior to ADASYN in terms of F1 score, it still outperforms SMOTE and Borderline-SMOTE. Therefore, based on its overall performance, the safe-level SMOTE algorithm is chosen as the optimal oversampling technique for this application.

E. Test III Diagnostic Model Performance

In the field of power electronics fault diagnosis, it is crucial to achieve high accuracy while minimizing computational complexity. Therefore, the selection of base classifiers with high accuracy and fast computation speed is essential for implementing ensemble learning algorithms. Thus, the performance of various classification models, including KNN, SVM [13], DT, RF [16], ELM [15], BPNN [37], CNN [20], and RVFL [17], is evaluated on a balanced dataset created with an imbalanced ratio of 4:1:1:1:1. It also should be noted that RVFLs represent the ensemble model.

To satisfy the needs of diagnostic model online computing with high accuracy, the parameters of the base classifier RVFL, as well as the ensemble size, are determined based on the computing resources available within the diagnostic platform. For a single RVFL model, the activation function and the hidden node number should be tuned by the historical data. With a compromised diagnosis accuracy, the activation function is selected as “sigmoid,” while the hidden layer number is 200, and n is 5 in this study. The evaluation includes metrics such as training accuracy, test accuracy, test time, F1 score, and G-mean, as presented in Table II. Note that the time is computed by MATLAB 2021b with Intel(R) Core(TM) i5-10210 U CPU at 1.6 GHz processor and 8-GB RAM. Moreover, their corresponding performance metrics for the raw data with different imbalanced ratios are presented in Table III.

As can be seen, both BPNN and RVFL exhibit comparable test time, while the CNN and RVFLs performs same train accuracy on the same test data. However, when considering multiple factors, RVFLs outperforms others in terms of test accuracy, F1 score, and G-mean. Specifically, RVFLs performs slightly better than them in G-mean, signifying its superior performance in handling imbalanced data. The single RVFL model is regarded as the fundamental ensemble model as it stands out with the

TABLE III
SINGLE AND ENSEMBLE MODEL COMPARISON UNDER IMBALANCED RATIOS
OF 8:1:1:1:1 AND 12:1:1:1:1

Imbalanced ratio	8:1:1:1:1		12:1:1:1:1	
Diagnostic model	RVFL	RVFLs	RVFL	RVFLs
Train accuracy	0.9970	1	0.9939	1
Test accuracy	0.8857	0.9143	0.8381	0.8476
Test time	0.0156	1.2656	0.0156	1.4375
F1 score	0.8503	0.8988	0.7214	0.7522
G-mean	0.8988	0.9103	0.8096	0.8223

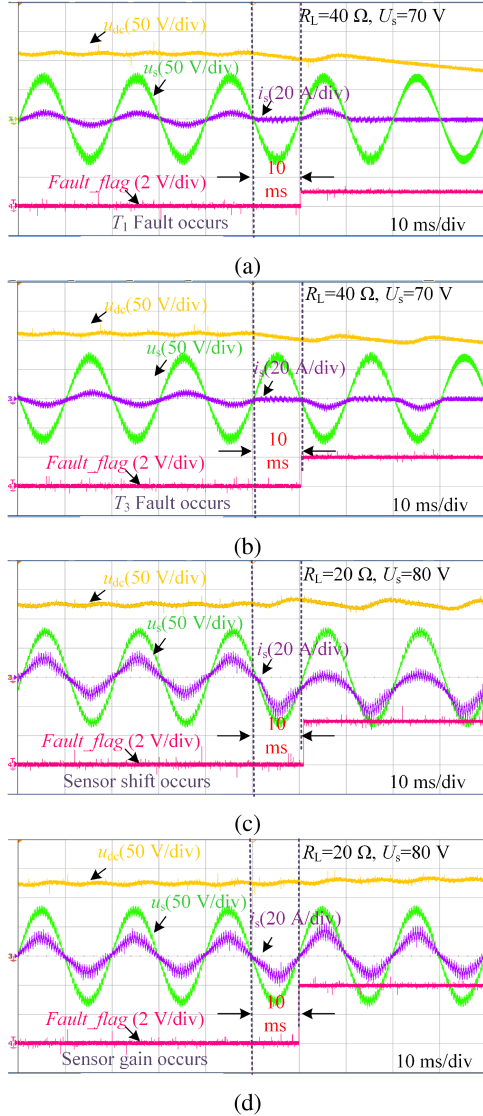


Fig. 12. Diagnostic results, including (a) T_1 open-circuit fault, (b) T_3 open-circuit fault, (c) sensor shift fault, and (d) sensor gain fault.

values for test time, F1 score, and G-mean, surpassing other classification models by 0.0156 s, 96%, and 97.7%, respectively. Moreover, the ensemble RVFLs model further enhances training accuracy, test accuracy, F1 score, and G-mean at the expense of increased test time. This tradeoff is well-suited for scenarios where diagnostic precision is of utmost importance.

F. Test IV Real-Time Diagnosis Implementation

The experimental platform was designed to demonstrate on-line fault diagnosis for a single-phase rectifier. The experimental setup is illustrated in Fig. 7 and the corresponding experimental results are presented in Fig. 12. The control and diagnosis algorithms run on an RT box controller, which utilizes a Xilinx Zynq Z-7030 system-on-chip with two CPU cores embedded in an field-programmable gate array. During the online application stage, the grid-side current signals are sampled at a frequency of 20 kHz. Since the fundamental period of grid-side current is 0.02 s, there are 200 data points collected per half period. Once the sampling is completed, the diagnostic program will start to identify faults.

Once the IGBT open-circuit faults and sensor faults are identified, a protection signal is generated to deactivate all IGBT control signals. Specifically, in the case of IGBT T_1 and T_3 open-circuit faults, with a load resistance R_L of 40 Ω and a grid-side voltage magnitude U_s of 70 V, the respective output fault flags are 1 and 2. On the other hand, the current sensor shift and gain faults are tested with an R_L of 20 Ω and a U_s of 80 V, resulting in fault flags of 3 and 4, respectively.

Thanks to the feature selection process, the computational complexity is significantly reduced, resulting in an online calculation time of less than 0.1 ms. This is negligible compared to the data sampling time of 10 ms. Consequently, all faults can be identified within approximately 10 ms using the proposed method. Moreover, the method is capable of adapting to various load conditions and variations in the grid-side voltage. Therefore, this effective implementation of online fault diagnosis further validates the effectiveness of the proposed scheme.

V. CONCLUSION

This article proposes a method integrating of oversampling and ensemble learning for fault diagnosis in IGBT open-circuit and sensor faults of single-phase PWM rectifier systems with imbalanced data. The work consists of the following three main aspects:

- 1) the hybrid feature selection scheme to obtain key and low-dimensional features;
- 2) the safe-level SMOTE algorithm for balancing the dataset;
- 3) the ensemble RVFLs model for enhancing diagnostic accuracy.

The proposed method is verified on an experimental dataset. Four experiments are conducted to evaluate the effectiveness of the selected features, the resampled dataset, the trained ensemble model, and the real-time diagnostic process. The proposed method is effective in addressing the fault diagnosis issue with the constraints of the imbalanced data, showcasing improvements in precise and demonstrating its suitability for real-time implementation in the application of the power electronic converter. Also, when applying this approach to other fault types and platforms, its applicability requires further exploration since the selection of sampling methods and diagnostic algorithms relies on the validation results from data acquired on the dedicated experimental platform.

REFERENCES

- [1] J. F. Martins, E. Romero-Cadaval, D. Vinnikov, and M. Malinowski, "Transactive energy: Power electronics challenges," *IEEE Power Electron. Mag.*, vol. 9, no. 1, pp. 20–32, Mar. 2022.
- [2] K. Fischer et al., "Reliability of power converters in wind turbines: Exploratory analysis of failure and operating data from a worldwide turbine fleet," *IEEE Trans. Power Electron.*, vol. 34, no. 7, pp. 6332–6344, Jul. 2019.
- [3] S. Zhao and H. Wang, "Enabling data-driven condition monitoring of power electronic systems with artificial intelligence: Concepts, tools, and developments," *IEEE Power Electron. Mag.*, vol. 8, no. 1, pp. 18–27, Mar. 2021.
- [4] H. Liu, Z. Liu, W. Jia, D. Zhang, and J. Tan, "A novel imbalanced data classification method based on weakly supervised learning for fault diagnosis," *IEEE Trans. Ind. Inform.*, vol. 18, no. 3, pp. 1583–1593, Mar. 2022.
- [5] Z. Gao, C. Cecati, and S. X. Ding, "A survey of fault diagnosis and fault-tolerant techniques—part II: Fault diagnosis with knowledge-based and hybrid/active approaches," *IEEE Trans. Ind. Electron.*, vol. 62, no. 6, pp. 3768–3774, Jun. 2015.
- [6] N. Wassinger, E. Penovi, R. G. Retegui, and S. Maestri, "Open-circuit fault identification method for interleaved converters based on time-domain analysis of the state observer residual," *IEEE Trans. Power Electron.*, vol. 34, no. 4, pp. 3740–3749, Apr. 2019.
- [7] D. Xie et al., "Diagnosis and resilient control for multiple sensor faults in cascaded h-bridge multilevel converters," *IEEE Trans. Power Electron.*, vol. 38, no. 9, pp. 11435–11450, Sep. 2023.
- [8] B. Gou, X. Ge, S. Wang, X. Feng, J. B. Kuo, and T. G. Habetler, "An open-switch fault diagnosis method for single-phase PWM rectifier using a model-based approach in high-speed railway electrical traction drive system," *IEEE Trans. Power Electron.*, vol. 31, no. 5, pp. 3816–3826, May 2016.
- [9] X. Ge, J. Pu, and Y. Liu, "Online open-switch fault diagnosis method in single-phase PWM rectifiers," *Electron. Lett.*, vol. 51, no. 23, pp. 1920–1922, Nov. 2015.
- [10] T. Kim, W. Lee, and D. Hyun, "Detection method for open-circuit fault in neutral-point-clamped inverter systems," *IEEE Trans. Ind. Electron.*, vol. 56, no. 7, pp. 2754–2763, Jul. 2009.
- [11] B. K. Bose, "Artificial intelligence techniques: How can it solve problems in power electronics?: An advancing frontier," *IEEE Power Electron. Mag.*, vol. 7, no. 4, pp. 19–27, Dec. 2020.
- [12] S. Zhao, F. Blaabjerg, and H. Wang, "An overview of artificial intelligence applications for power electronics," *IEEE Trans. Power Electron.*, vol. 36, no. 4, pp. 4633–4658, Apr. 2021.
- [13] I. Bandyopadhyay, P. Purkait, and C. Koley, "Performance of a classifier based on time-domain features for incipient fault detection in inverter drives," *IEEE Trans. Ind. Inform.*, vol. 15, no. 1, pp. 3–14, Jan. 2019.
- [14] B. Cai, Y. Zhao, H. Liu, and M. Xie, "A data-driven fault diagnosis methodology in three-phase inverters for PMSM drive systems," *IEEE Trans. Power Electron.*, vol. 32, no. 7, pp. 5590–5600, Jul. 2017.
- [15] B. Gou, Y. Xu, Y. Xia, G. Wilson, and S. Liu, "An intelligent time-adaptive data-driven method for sensor fault diagnosis in induction motor drive system," *IEEE Trans. Ind. Electron.*, vol. 66, no. 12, pp. 9817–9827, Dec. 2019.
- [16] L. Kou, C. Liu, G. Cai, J. Zhou, and Q. Yuan, "Data-driven design of fault diagnosis for three-phase PWM rectifier using random forests technique with transient synthetic features," *IET Power Electron.*, vol. 13, no. 16, pp. 3571–3579, Dec. 2020.
- [17] B. Gou, Y. Xu, Y. Xia, Q. Deng, and X. Ge, "An online data-driven method for simultaneous diagnosis of IGBT and current sensor fault of three-phase PWM inverter in induction motor drives," *IEEE Trans. Power Electron.*, vol. 35, no. 12, pp. 13281–13294, Dec. 2020.
- [18] Q. Deng, B. Gou, X. Ge, C. Lin, D. Xie, and X. Feng, "A high-accuracy-light-AI data-driven diagnosis method for open-circuit faults in single-phase PWM rectifiers," *IEEE Trans. Transp. Electrification*, vol. 9, no. 3, pp. 4352–4365, Sep. 2023.
- [19] L. Xu, M. Cao, B. Song, J. Zhang, Y. Liu, and F. E. Alsaadi, "Open-circuit fault diagnosis of power rectifier using sparse autoencoder based deep neural network," *Neurocomputing*, vol. 311, pp. 1–10, May 2018.
- [20] L. Wen, X. Li, L. Gao, and Y. Zhang, "A new convolutional neural network based data-driven fault diagnosis method," *IEEE Trans. Ind. Electron.*, vol. 65, no. 7, pp. 5990–5998, Jul. 2018.
- [21] Z. Liu, L. Fang, D. Jiang, and R. Qu, "A machine-learning-based fault diagnosis method with adaptive secondary sampling for multiphase drive systems," *IEEE Trans. Power Electron.*, vol. 37, no. 8, pp. 8767–8772, Aug. 2022.
- [22] D. Chicco and L. Oneto, "An enhanced random forests approach to predict heart failure from small imbalanced gene expression data," *IEEE-ACM Trans. Comput. Biol. Bioinform.*, vol. 18, no. 6, pp. 2759–2765, Nov./Dec. 2021.
- [23] L. Chen, S. Wan, and L. Dou, "Improving diagnostic performance of high-voltage circuit breakers on imbalanced data using an oversampling method," *IEEE Trans. Power Del.*, vol. 37, no. 4, pp. 2704–2716, Aug. 2022.
- [24] J. Zhang, K. Zhang, Y. An, H. Luo, and S. Yin, "An integrated multitasking intelligent bearing fault diagnosis scheme based on representation learning under imbalanced sample condition," *IEEE Trans. Neural Netw. Learn. Syst.*, to be published, doi: [10.1109/TNNLS.2022.3232147](https://doi.org/10.1109/TNNLS.2022.3232147).
- [25] A. Ito, K. Saito, R. Ueno, and N. Homma, "Imbalanced data problems in deep learning-based side-channel attacks: Analysis and solution," *IEEE Trans. Inf. Forensics Secur.*, vol. 16, pp. 3790–3802, Jun. 2021.
- [26] A. Onan, "Consensus clustering-based undersampling approach to imbalanced learning," *Sci. Prog.*, vol. 2019, Mar. 2019.
- [27] F. Jia, S. Li, H. Zuo, and J. Shen, "Deep neural network ensemble for the intelligent fault diagnosis of machines under imbalanced data," *IEEE Access*, vol. 8, pp. 120974–120982, 2020.
- [28] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, Jun. 2002.
- [29] C. Bunkhumpornpat, K. Sinapiromsaran, and C. Lursinsap, "Safe-level-SMOTE: Safe-level-synthetic minority over-sampling technique for handling the class imbalanced problem," in *Proc. Pacific-Asia Conf. Knowl. Discov. Data Mining*, 2009, pp. 475–482.
- [30] H. Han, W.-Y. Wang, and B.-H. Mao, "Borderline-SMOTE: A new over-sampling method in imbalanced data sets learning," in *Proc. Int. Conf. Intell. Comput.*, 2005, pp. 878–887.
- [31] H. He, Y. Bai, E. A. Garcia, and S. Li, "ADASYN: Adaptive synthetic sampling approach for imbalanced learning," in *Proc. IEEE Int. Joint Conf. Neural Netw.*, 2008, pp. 1322–1328.
- [32] S. J. Liu, "Statistical and analysis of traction system faults for electric multiple unit," *Railw. Loco. Car.*, vol. 33, no. 5, pp. 80–85, 2013.
- [33] A. Onan and S. Korukoğlu, "A feature selection model based on genetic rank aggregation for text sentiment classification," *J. Inf. Sci.*, vol. 43, no. 1, pp. 25–38, Feb. 2017.
- [34] H. Peng, F. Long, and C. Ding, "Feature selection based on mutual information: Criteria of max-dependency, max-relevance, and min-redundancy," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 8, pp. 1226–1238, Aug. 2005.
- [35] J. Shi et al., "Cascaded multi-column RVFL classifier for single-modal neuroimaging-based diagnosis of Parkinson's disease," *IEEE Trans. Biomed. Eng.*, vol. 66, no. 8, pp. 2362–2371, Aug. 2019.
- [36] Y.-H. Pao and Y. Takefuji, "Functional-link net computing: Theory system architecture and functionalities," *Computer*, vol. 25, no. 5, pp. 76–79, May 1992.
- [37] J. Zhang et al., "Lifetime extension approach based on the Levenberg–Marquardt neural network and power routing of DC–DC converters," *IEEE Trans. Power Electron.*, vol. 38, no. 8, pp. 10280–10291, Aug. 2023.



Qingli Deng (Student Member, IEEE) received the B.S. degree in electrical engineering and automation from Southwest Petroleum University, Chengdu, China, in 2018. She is currently working toward the Ph.D. degree in electrical engineering with Southwest Jiaotong University, Chengdu.

Her research interests include fault diagnosis and fault-tolerant control of traction converter systems, along with the intelligent application of artificial intelligence in power electronic systems.



Xiaoyun Feng received the B.S., M.S., and Ph.D. degrees in electrical engineering from Southwest Jiaotong University, Chengdu, China, in 1983, 1988 and 2001, respectively.

From 1998 to 1999, she was a Visiting Professor with the University of Tokyo. Since 1983, she has been with the College of Electrical Engineering, Southwest Jiaotong University, where she is currently a Full Professor. Her main research interests include railway optimizing operation, real-time control of train operation, energy-efficient driving, the design of driver assistant systems, the analysis and control of electrical traction converter, and motor drive systems.



Shuai Zhao (Member, IEEE) received the B.S. (Hons.), M.S., and Ph.D. degrees in information and telecommunication engineering from Northwestern Polytechnical University, Xi'an, China, in 2011, 2014, and 2018, respectively.

From 2014 to 2016, he was a Visiting Ph.D. student with the University of Toronto, Toronto, Canada. In August 2018, he was a Visiting Scholar with the University of Texas at Dallas, USA. From 2018 to 2022, he was a Postdoc Researcher with AAU Energy, Aalborg University, Aalborg, Denmark. He is currently an Assistant Professor with AAU Energy, Aalborg University. His research interests include physics-informed machine learning, system informatics, condition monitoring, diagnostics and prognostics, and tailored artificial intelligence tools for power electronic systems.

He is an Associate Editor for IEEE TRANSACTIONS ON VEHICLE TECHNOLOGY and a Guest Editor for IEEE JOURNAL OF EMERGING AND SELECTED TOPICS IN INDUSTRIAL ELECTRONICS and *Elsevier e-Prime*.



Bin Gou (Member, IEEE) received the B.S. and Ph.D. degrees in electrical engineering from Southwest Jiaotong University, Chengdu, China, in 2010 and 2016.

From 2014 to 2015, he was a Visiting Student with the School of Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA, USA. From 2017 to 2020, he was a Research Fellow with the Rolls-Royce @ NTU Corporate Laboratory, Nanyang Technological University, Singapore. He is currently with the School of Electrical Engineering, Southwest Jiaotong University. His research interest includes data-analytics for fault diagnosis and health management of power drive and energy storage system.



Huai Wang (Senior Member, IEEE) received the B.E. degree in electrical engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2007, and the Ph.D. degree in power electronics from the City University of Hong Kong, Hong Kong, in 2012.

He was a Visiting Scientist with the ETH Zurich, Switzerland, from August to September 2014, and with the Massachusetts Institute of Technology (MIT), USA, from September to November 2013.

He was with the ABB Corporate Research Center, Switzerland, in 2009. He is currently a Professor with AAU Energy, Aalborg University, Aalborg, Denmark, where he leads the group of Reliability of Power Electronic Converters (ReliaPEC) and the mission on Digital Transformation and AI. His research interests include the fundamental challenges in modeling and validating power electronic component failure mechanisms and application issues in system-level predictability, condition monitoring, circuit architecture, and robustness design.

Dr. Wang was the recipient the Richard M. Bass Outstanding Young Power Electronics Engineer Award from the IEEE Power Electronics Society in 2016 and the first Prize Paper Award from IEEE TRANSACTIONS ON POWER ELECTRONICS in 2021. He serves as an Associate Editor for *Journal of Emerging and Selected Topics in Power Electronics* and IEEE TRANSACTIONS ON POWER ELECTRONIC. He was elected as a Member of the Danish Academy of Technical Sciences in 2023.



Xinglai Ge (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering from Southwest Jiaotong University, Chengdu, China, in 2001, 2004, and 2010, respectively.

From July to August of 2012, he was a Visiting Scholar with George Mason University, Fairfax, VA, USA. From October 2013 to 2014, he was a Visiting Scholar with the School of Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA, USA. He is currently a Full Professor with the School of Electrical Engineering, Southwest Jiaotong University and the Vice Director with the Department of Power Electronics and Power Drive.

He is the author and co-author of more than 60 technical papers. His research interests include stability analysis and control of electrical traction system, fault diagnosis and hardware-in-the-loop simulation of traction converter, and motor drive system.